3.33

$$\frac{d}{dt}y(t) + 4y(t) = x(t)$$

input $x(t)$ output $y(t)$

The frequency response of the system $H(j\omega)$ input as $x(t) = e^{j\omega t}$

$$y(t) = H(j\omega)e^{j\omega t}$$

$$\frac{d}{dt}\left[H(j\omega)e^{j\omega t}\right] + 4\left[H(j\omega)e^{j\omega t}\right] = e^{j\omega t}$$

$$j\omega H(j\omega)e^{j\omega t} + 4H(j\omega)e^{j\omega t} = e^{j\omega t}$$

$$H(j\omega)e^{j\omega t}(j\omega + 4) = e^{j\omega t}$$

$$H(j\omega) = \frac{1}{j\omega + 4}$$

$$y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0} \rightarrow y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jk\omega_0}$$

$x(t)$ => input

a_k is Fourier series coefficient of $x(t)$ [input signal]

ω_0 is fundamental frequency

$b_k = a_k H(jk\omega_0)$ is the Fourier series coefficient of $y(t)$

a]

input $x(t) = \cos 2\pi t$ $\quad \omega_0 = 2\pi$

$$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$$

$$x(t) = \frac{e^{j2\pi t} + e^{-j2\pi t}}{2}$$

$$= \frac{1}{2}e^{j2\pi t} + \frac{1}{2}e^{-j2\pi t}$$

Fourier series coefficient of $x(t)$

$a_1 = \frac{1}{2}$ $\quad a_{-1} = \frac{1}{2}$

$$\begin{cases} a_k = \frac{1}{2}, & k = 1, -1 \\ a_k = 0, & \text{otherwise} \end{cases}$$

$$b_k = a_k H(jk\omega_0)$$

$$= a_k\left[\frac{1}{jk\omega_0 + 4}\right]$$

$$b_1 = a_1\left[\frac{1}{j\omega_0 + 4}\right]$$

$a_1 = \frac{1}{2}$

$\omega_0 = 2\pi$

$$b_1 = \frac{1}{2(4+j2\pi)} = \frac{1}{4(2+j\pi)}$$

$$b_{-1} = a_{-1}\left[\frac{1}{-jw_0+4}\right] \qquad a_{-1} = \frac{1}{2} \qquad w_0 = 2\pi$$

$$b_{-1} = \frac{1}{2(4-j2\pi)} = \frac{1}{4(2-j\pi)}$$

the polar form of $\frac{1}{a+jb}$

$$\frac{1}{a+jb} = \frac{1}{\sqrt{a^2+b^2}} e^{-j\tan^{-1}\left(\frac{b}{a}\right)}$$

$a=2$, $b=-\pi$

$$b_1 = \frac{1}{4\sqrt{4+\pi^2}} e^{-j\tan^{-1}\left(\frac{-\pi}{2}\right)} \rightarrow \frac{1}{4\sqrt{4+\pi^2}} e^{+j\tan^{-1}\left(\frac{\pi}{2}\right)}$$

$$b_1 = \frac{1}{4\sqrt{4+\pi^2}} e^{-j\tan^{-1}\left(\frac{\pi}{2}\right)}$$

$$\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$$

$$y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jkw_0 t}$$

$$= b_{-1}e^{-j2\pi t} + b_1 e^{j2\pi t} = \frac{1}{4\sqrt{4+\pi^2}} e^{j\tan^{-1}\left(\frac{\pi}{2}\right)} e^{-j2\pi t} + \frac{1}{4\sqrt{4+\pi^2}} e^{-j\tan^{-1}\left(\frac{\pi}{2}\right)} e^{j2\pi t}$$

$$y(t) = \frac{1}{2\sqrt{4+\pi^2}}\left[\frac{e^{j\left(\tan^{-1}\left(\frac{\pi}{2}\right)-2\pi t\right)} + e^{-j\left(2\pi t-\tan^{-1}\left(\frac{\pi}{2}\right)\right)}}{2}\right]$$

$$y(t) = \frac{1}{2\sqrt{4+\pi^2}}\left[\cos\left(\tan^{-1}\left(\frac{\pi}{2}\right)-2\pi t\right)\right]$$

b) $x(t) = \sin 4\pi t + \cos(6\pi t + \pi/4)$

$= \frac{1}{2j}\left[e^{j4\pi t} - e^{-j4\pi t}\right] + \frac{e^{j(6\pi t + \frac{\pi}{4})} + e^{-j(6\pi t + \frac{\pi}{4})}}{2}$

$= \frac{1}{2j} e^{j4\pi t} - \frac{1}{2j} e^{-j4\pi t} + \frac{1}{2} e^{j(6\pi t + \frac{\pi}{4})} + \frac{1}{2} e^{-j(6\pi t + \pi/4)}$

$= \frac{1}{2j} e^{j2\pi t/2} - \frac{1}{2j} e^{-(j2\pi t)2} + \frac{e^{j\frac{\pi}{4}}}{2} e^{(j2\pi t)3} - \frac{e^{-j\frac{\pi}{4}}}{2} e^{-(j2\pi t)3}$

w_0 = greatest common vector $(4\pi, 6\pi)$

$w_0 = 2\pi$

$\cos\theta = \frac{e^{j\theta} + e^{-j\theta}}{2}$

$\sin\theta = \frac{1}{2j}\left[e^{j\theta} - e^{-j\theta}\right]$

Thus, the fourier series coefficients of $x(t)$;

$a_2 = \frac{1}{2j}$ $\quad a_{-2} = \frac{-1}{2j}$

$a_3 = \frac{1}{2} e^{j\pi/4}$ $\quad a_{-3} = \frac{-1}{2} e^{-j\pi/4}$

$\Rightarrow b_k = a_k / (jk\omega_0)$

$= a_k \left[\frac{1}{4 + jk\omega_0}\right]$

$k = 2$

$b_2 = \frac{1}{2j}\left[\frac{1}{4 + j4\pi}\right] = \frac{1}{8}\left[\frac{1}{j - \pi}\right]$

$j^2 = -1$

$b_{-2} = \frac{-1}{2j}\left[\frac{1}{4 + j4\pi}\right] = \frac{1}{8}\left[\frac{1}{-j - \pi}\right]$

$b_3 = \frac{1}{2} e^{j\pi/4}\left[\frac{1}{4 + j6\pi}\right] = \frac{1}{4}\left[\frac{e^{j\pi/4}}{2 + 3j\pi}\right]$

$b_{-3} = \frac{1}{2} e^{-j\frac{\pi}{4}}\left[\frac{1}{4 - j6\pi}\right] = \frac{1}{4}\left[\frac{e^{-j\pi/4}}{2 - j3\pi}\right]$

So, the Fourier series coefficient of $y(t)$

$$b_2 = \frac{1}{8(-\pi + j)}, \qquad b_{-2} = \frac{1}{8(-\pi - j)}$$

$$b_3 = \frac{1}{4}\left[\frac{e^{j\pi/4}}{2 + j3\pi}\right] \qquad b_{-3} = \frac{1}{4}\left[\frac{e^{-j\pi/4}}{2 - j3\pi}\right]$$

$$\begin{cases} \tan^{-1}\left(\frac{b}{-a}\right) = \pi - \tan^{-1}\left(\frac{b}{a}\right) \\ \tan^{-1}\left(\frac{-b}{-a}\right) = -\pi + \tan^{-1}\left(\frac{b}{a}\right) \\ \tan^{-1}\left(\frac{-b}{a}\right) = -\tan^{-1}\left(\frac{b}{a}\right) \end{cases}$$

We should convert to polar forms of b_2, b_3, b_{-2} and b_{-3}

$$b_2 = \frac{1}{8\sqrt{1+\pi^2}} e^{-j\tan^{-1}\left(\frac{1}{-\pi}\right)} = \frac{1}{8\sqrt{1+\pi^2}} e^{-j\left(\pi - \tan^{-1}\left(\frac{1}{\pi}\right)\right)}$$

$$b_{-2} = \frac{1}{8\sqrt{1+\pi^2}} e^{-j\tan^{-1}\left(\frac{-1}{-\pi}\right)} = \frac{1}{8\sqrt{1+\pi^2}} e^{-j\left(-\pi + \tan^{-1}\left(\frac{1}{\pi}\right)\right)}$$

$$b_3 = \frac{e^{j\frac{\pi}{4}}}{4\sqrt{4+9\pi^2}} e^{-j\tan^{-1}\left(\frac{3\pi}{2}\right)}$$

$$b_{-3} = \frac{e^{-j\pi/4}}{4\sqrt{4+9\pi^2}} e^{-j\tan^{-1}\left(\frac{-3\pi}{2}\right)} = \frac{e^{-j\frac{\pi}{4}}}{8\sqrt{1+\pi^2}} e^{j\tan^{-1}\left(\frac{3\pi}{2}\right)}$$

$$y(t) = \sum_{k=-\infty}^{\infty} b_k e^{jk2\pi t}$$

$$= b_{-3}e^{-j6\pi t} + b_{-2}e^{-j4\pi t} + b_2 e^{j4\pi t} + b_3 e^{j6\pi t}$$

$$= \frac{1}{4\sqrt{1+\pi^2}}\left[\frac{e^{j\left(4\pi t - \pi + \tan^{-1}\left(\frac{1}{\pi}\right)\right)} + e^{-j\left(4\pi t - \pi + \tan^{-1}\left(\frac{1}{\pi}\right)\right)}}{2} + \frac{e^{j\left(6\pi t + \frac{\pi}{4} - \tan^{-1}\left(\frac{3\pi}{2}\right)\right)} + e^{-j\left(6\pi t + \frac{\pi}{4} - \tan^{-1}\left(\frac{3\pi}{2}\right)\right)}}{2}\right]$$

result is

$$\frac{1}{4\sqrt{1+\pi^2}}\left[\cos\left(4\pi t - \pi + \tan^{-1}\left(\frac{1}{\pi}\right)\right) + \cos\left(6\pi t + \frac{\pi}{4} - \tan^{-1}\left(\frac{3\pi}{2}\right)\right)\right]$$

3.36 $y[n] - \frac{1}{4}y[n-1] = x[n]$

$Y(e^{j\omega}) - \frac{1}{4}e^{-j\omega}Y(e^{j\omega}) = X(e^{j\omega})$

$\left(1 - \frac{1}{4}e^{-j\omega}\right)Y(e^{j\omega}) = X(e^{j\omega})$

$\frac{Y(e^{j\omega})}{X(e^{j\omega})} = \frac{1}{\left(1-\frac{1}{4}e^{-j\omega}\right)}$ $\qquad H(e^{j\omega}) = \frac{1}{\left(1-\frac{1}{4}e^{-j\omega}\right)}$

* $x[n] = \sum_{k=\langle N\rangle} a_k e^{jk(2\pi/N)n}$ $\qquad$ input $\Rightarrow x[n]$ $\qquad$ impulse response $h[n]$

output $\Rightarrow y[n]$

a)

$x[n] = \sin\left(\frac{3\pi}{4}n\right)$

$= \sin\left(\frac{2\pi}{8}3n\right) = \frac{e^{j\frac{2\pi}{8}3n} - e^{-j\frac{2\pi}{8}3n}}{2j}$ $\qquad N = 8$

$x[n] = \frac{1}{2j}e^{j\frac{2\pi}{8}3n} + \left(\frac{-1}{2j}\right)e^{-j\frac{2\pi}{8}3n}$ $\qquad a_3 = \frac{1}{2j}$, $a_{-3} = \frac{-1}{2j}$

general equation $x[n] = \sum_{\langle N\rangle} a_k e^{jk\omega_0 n} \Rightarrow \sum_{\langle N\rangle} a_k e^{jk\left(\frac{2\pi}{N}\right)n}$

$y[n] = \sum_{k=8} a_k H\left(e^{jk\left(\frac{2\pi}{8}\right)}\right) e^{jk\left(\frac{2\pi}{8}\right)n}$

$= a_{-3}H\left(e^{j(-3)\left(\frac{2\pi}{8}\right)}\right)e^{j(-3)\left(\frac{2\pi}{8}\right)n} + a_3 H\left(e^{j(3)\left(\frac{2\pi}{8}\right)}\right)e^{j(3)\left(\frac{2\pi}{8}\right)n}$

$= \left[\left(\frac{-1}{2j}\right)\left(\frac{1}{\left(1-\frac{1}{4}e^{j\left(\frac{3\pi}{4}\right)}\right)}\right)\right]e^{-j\left(\frac{3\pi}{4}\right)n} + \left[\left(\frac{1}{2j}\right)\left(\frac{1}{\left(1-\frac{1}{4}e^{j\left(\frac{3\pi}{4}\right)}\right)}\right)\right]e^{j\left(\frac{3\pi}{4}\right)n}$

b) $x[n] = \cos\left(\frac{\pi}{4}n\right) + 2\cos\left(\frac{\pi}{2}n\right)$

$= \cos\left(\frac{2\pi}{8}n\right) + 2\cos\left(\frac{2\pi}{8}2n\right)$

$= \frac{e^{j\frac{2\pi}{8}n} + e^{-j\frac{2\pi}{8}n}}{2} + 2\left[e^{j\frac{2\pi}{8}(2n)} + e^{-j\frac{2\pi}{8}(2n)}\right]$

$x[n] = e^{-j\frac{2\pi}{8}(2n)} + \frac{1}{2}e^{-j\frac{2\pi}{8}n} + \frac{1}{2}e^{j\frac{2\pi}{8}n} + e^{j\frac{2\pi}{8}(2n)}$ $\rightarrow a_1 = a_{-1} = \frac{1}{2}$, $a_2 = a_{-2} = 1$

general equation $\Rightarrow$ $x[n] = \sum_{\langle N \rangle} a_k e^{jk\omega_0 n} \Rightarrow \sum_{N} a_k e^{jk\left(\frac{2\pi}{N}\right)n}$

$N=8$

$y[n] = \sum_{k=8} a_k H\left(e^{jk\left(\frac{2\pi}{8}\right)}\right) e^{jk\left(\frac{2\pi}{8}\right)n}$

$= \left[a_{-2} H\left[e^{j(-2)\left(\frac{2\pi}{8}\right)}\right] e^{j(-2)\left(\frac{2\pi}{8}\right)n} + a_{-1} H\left(e^{j(-1)\left(\frac{2\pi}{8}\right)}\right) e^{j(-1)\left(\frac{2\pi}{8}\right)n} + \right.$

$\left. a_1 H\left(e^{j(1)\left(\frac{2\pi}{8}\right)}\right) e^{j(1)\left(\frac{2\pi}{8}\right)n} + a_2 H\left(e^{j(2)\left(\frac{2\pi}{8}\right)}\right) e^{j(2)\left(\frac{2\pi}{8}\right)n}\right]$

$= \left[(1)\frac{1}{\left(1-\frac{1}{4}e^{-j(-2)\left(\frac{2\pi}{8}\right)}\right)} e^{j(-2)\left(\frac{2\pi}{8}\right)n} + \left(\frac{1}{2}\right)\frac{1}{\left(1-\frac{1}{4}e^{-j(-1)\left(\frac{2\pi}{8}\right)}\right)} e^{j(-1)\left(\frac{2\pi}{8}\right)n} \right.$

$\left. + \left(\frac{1}{2}\right)\frac{1}{\left(1-\frac{1}{4}e^{-j(1)\left(\frac{2\pi}{8}\right)}\right)} e^{j(1)\left(\frac{2\pi}{8}\right)n} + (1)\frac{1}{\left(1-\frac{1}{4}e^{-j(2)\left(\frac{2\pi}{8}\right)}\right)} e^{j(2)\left(\frac{2\pi}{8}\right)n}\right]$

result is

$y[n] = \frac{1}{\left(1-\frac{1}{4}e^{-j\left(\frac{\pi}{2}\right)}\right)} e^{-j\left(\frac{\pi}{2}\right)n} + \left[\frac{1}{2}\frac{1}{\left(1-\frac{1}{4}e^{-j\left(\frac{\pi}{4}\right)}\right)}\right] e^{-j\left(\frac{\pi}{4}\right)n} + \left[\frac{1}{2}\frac{1}{\left(1-\frac{1}{4}e^{-j\left(\frac{\pi}{4}\right)}\right)}\right] e^{j\left(\frac{\pi}{4}\right)n}$

$+ \left[\frac{1}{\left(1-\frac{1}{4}e^{-j\left(\frac{\pi}{2}\right)}\right)}\right] e^{j\left(\frac{\pi}{2}\right)n}$